

Skills Summary

Legs, Hypotenuses, and the Pythagorean Theorem

Use this reference while completing your worksheet or when studying.

Skill 1 — Finding the hypotenuse: add the squares

The theorem: $a^2 + b^2 = c^2$

c is the **hypotenuse**: the side opposite the right angle, and always the longest side of the triangle. a and b are the two **legs**, the sides that meet at the right angle.

When both legs are known and the hypotenuse is missing: $c = \sqrt{a^2 + b^2}$.

Example: A right triangle has legs 6 and 8. Find the hypotenuse.

Step 1. Both known sides meet at the right angle, so they are legs and the missing side is the hypotenuse — add.

Step 2. $c = \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} \Rightarrow c = 10$

Try it: A right triangle has legs 5 and 12. Find the hypotenuse.

check: 13

Skill 2 — Finding a leg: subtract the squares

Rearranged: $a^2 = c^2 - b^2$, so $a = \sqrt{c^2 - b^2}$

Use this whenever the *longest* side is one of the numbers you were given. The hypotenuse's square always comes first, and the known leg's square is taken away from it. Getting this backwards is the single most common Pythagorean mistake.

Example: Hypotenuse 10, one leg 6. Find the other leg.

Step 1. The longest side is given, so the missing side is a leg — subtract, do not add.

Step 2. $a = \sqrt{10^2 - 6^2} = \sqrt{100 - 36} = \sqrt{64} \Rightarrow a = 8$

Try it: Hypotenuse 26, one leg 24. Find the other leg.

check: 10

Skill 3 — Checking someone else's substitution

Two questions, in this order:

- Which given side is the *longest*? That one is the hypotenuse, and it is the only side allowed on its own in $a^2 + b^2 = c^2$.
- Is the answer smaller than the hypotenuse? A leg must be. An answer bigger than the hypotenuse is wrong before you check the arithmetic.

Example: Hypotenuse 20, leg 16. A student wrote $\sqrt{20^2 + 16^2} = 25.61$. Correct it.

Step 1. The student's answer is bigger than the hypotenuse 20, which is impossible — so a leg was substituted as if it were the hypotenuse.

Step 2. The missing side is a leg: $\sqrt{20^2 - 16^2} = \sqrt{400 - 256} = \sqrt{144} \Rightarrow$ **12**

Try it: Hypotenuse 41, leg 40. A student wrote $\sqrt{41^2 + 40^2} = 57.28$. Correct it.

6 :check

(!) Watch out: on a coordinate grid the horizontal and vertical gaps are the legs and the segment itself is the hypotenuse — so a distance is always found by *adding* the squares.